Cletus — Section 1, Day 3

- **Date:** [YYYY-MM-DD]
- **Course:** Week 1, Day 3 — Types of LLM, frontier models' strengths and pitfalls, agentic AI
- **Lab notebook:** none (lecture day, practised in chat apps)
- **Time spent:** [e.g. 1h]

Key concepts

- **Base model** — trained on huge amounts of text to predict the next token. It continues text but doesn't follow instructions well.
- **Chat (instruct) model** — a base model fine-tuned on conversations and human feedback (RLHF) so it follows instructions and chats. ChatGPT uses this kind of model.
- **Reasoning model** — trained to "think" step by step before answering, using hidden thinking tokens. Better at maths, logic and code, but slower and costs more. Examples: GPT-5 Thinking, DeepSeek R1.
- **The frontier models** — GPT (OpenAI), Claude (Anthropic), Gemini (Google), Grok (xAI), plus DeepSeek (open-weight).
- **Strengths** — summarizing and combining information, turning an outline into a full draft, writing and explaining code.
- **Pitfalls:**
- *Knowledge cutoff:* they don't know recent events unless they can search the web.
- *Niche or specialized topics:* answers can be shallow or wrong.
- *Hallucinations:* they can be confidently wrong.
- *Token quirks:* they see tokens, not letters, so counting letters is hard for them.
- **Agentic AI** — the model plans and takes several steps using tools:
- *Deep Research* searches the web and writes a report.
- *Claude Code* writes and runs code in your terminal.
- *Agent Mode* in ChatGPT uses a browser to complete tasks.
- **LLM competition game** — models play against each other, which shows how differently they behave. It also previews several LLMs talking to each other (Week 2).

Code / examples

No code today. Test prompts to compare models:

Test	Prompt	What it shows
Token quirk	How many times does the letter "r" appear in "strawberry"?	Models see tokens, not letters
Knowledge cutoff	What were the biggest AI announcements this week?	Cutoff vs web search
Reasoning	A short logic puzzle or riddle	Chat vs reasoning model
Hallucination	Summarize the paper "[made-up title]" by [made-up author]	Does it invent an answer or admit it doesn't know?

Exercise / challenge

1. Run the 4 prompts above in 2–3 free chat apps (ChatGPT, Claude, Gemini, DeepSeek).
2. Turn "thinking" or "reasoning" on and off where the app allows it, and compare.
3. Note in a small table which model was right, fastest and clearest.

Questions & blockers

Watch out for: - Don't trust answers about recent events or niche facts without checking them. - A better model isn't always the right choice: reasoning models cost more and are slower.

To discuss with my partner

- Which model won on which prompt? Did we get the same results?
- When is a reasoning model worth the extra time and cost?

Resources

- ChatGPT: https://chatgpt.com · Claude: https://claude.ai · Gemini: https://gemini.google.com · DeepSeek: https://chat.deepseek.com
- Compare models on a leaderboard: https://lmarena.ai